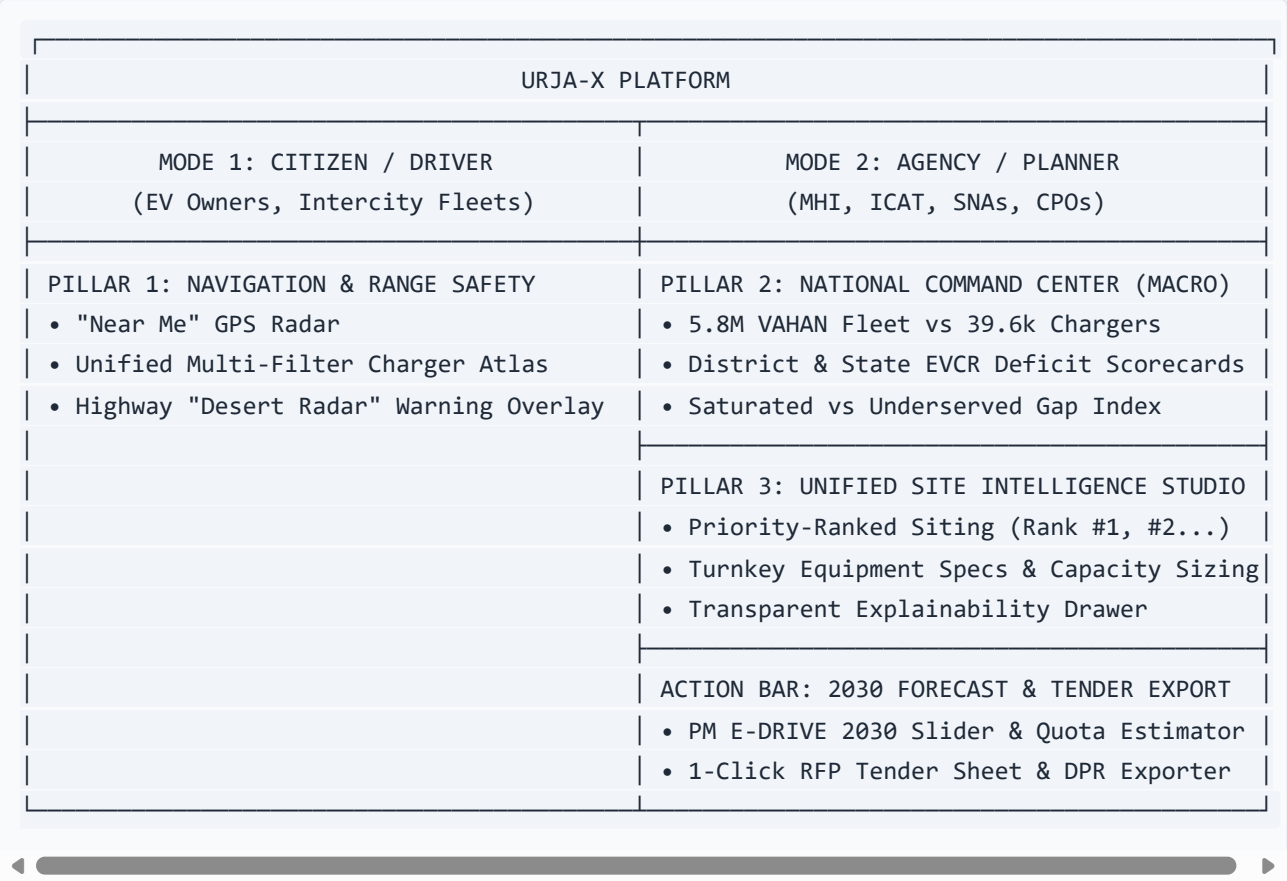
Urja-X: Simplified MVP Product Specification

National Geospatial Intelligence & Data-Driven Decision Support Platform for EV Infrastructure
Aligned with Ministry of Heavy Industries (MHI), ICAT, and Techfest 2026-27 Guidelines

---

1. Product Overview & Dual-Persona Architecture

Urja-X streamlines national EV charging planning into **three core functional pillars** serving two distinct user personas through a simple toggle switch:



---

2. Pillar 1: Citizen & EV Driver Mode (Navigation & Range Safety)

Designed for everyday EV owners, commercial cab fleets, and intercity travelers.

2.1 Unified EV Charger Map with "Near Me" GPS Radar

- **"Near Me" GPS Locator:** One-click GPS toggle automatically centers on the driver's current position and returns operational public chargers sorted by driving distance.
- **Granular Multi-Filters:**
- **Charging Speed & Power:** DC Ultra-Fast ($>120\text{ kW}$), DC Fast ($30\text{–}60\text{ kW}$), AC Fast ($11\text{–}22\text{ kW}$), AC Standard ($3.3\text{–}7.4\text{ kW}$).
- **Connector Standard:** CCS-2, Bharat DC-001, Type-2 AC, CHAdeMO.
- **Number of Charging Guns:** Filter for stations with ≥ 2 or ≥ 4 guns to avoid long wait queues.
- **CPO / Network Brand:** Tata Power, Jio-bp, Statiq, Ather Grid, Zeon, PSU oil pumps (IOCL, BPCL, HPCL).
- **Operational Status & Amenities:** 24/7 Access, Food/Café, Restrooms, Covered Canopy.

2.2 Highway "Desert Radar" for Safe Route Planning

- **Visual Highway Blackspot Overlay:** Renders colored risk segments along major expressways and National Highways.
- **Driver Range Warning:** Highlights highway stretches where **no public chargers exist for $>50\text{ km}$ or $>75\text{ km}$**, enabling intercity drivers to plan battery top-ups before entering high-risk charging deserts.

---

3. Pillar 2: National Command Center (Macro Policy & Regional Analytics)

Designed for the Ministry of Heavy Industries (MHI), ICAT, and State Nodal Agencies (SNAs).

3.1 Pan-India VAHAN vs. Charging Availability Map

- Interactive national choropleth mapping **5.8M+ VAHAN registered electric vehicles** against **39,641 operational public chargers** across all 36 States/UTs.
- Category-level breakdown: 2-Wheelers, 3-Wheelers, 4-Wheelers, and Electric Commercial Buses.

3.2 District & State Deficit Scorecards (EVCR Index)

- Real-time **EV-to-Charger Ratio (EVCR)** benchmarking against the international **IEA standard (10–20:1)**:
- 🔴 **Critical Deficit ($\text{EVCR} > 150$):** High vehicle adoption, acute infrastructure shortage.
- 🟡 **Transitional ($50 < \text{EVCR} \le 150$):** Moderate provisioning requiring expansion.

- 🟢 **Adequate ($\text{EVCR} \le 50$):** Well-provisioned for current fleet volume.
- **Target Charger Allocation ($\Delta N_{\text{chargers}}$):** Automatically computes the number of additional public charging stations required in each district to achieve policy parity ($\text{EVCR}_{\text{target}} = 15$).

---

4. Pillar 3: Unified Site Intelligence Studio (Micro-Planning & Turnkey Specs)

*This is the core engine of Urja-X. It unifies **Site Siting**, **Engineering Capacity Sizing**, and **Transparent Explainability** into one seamless, interactive workflow.*

Inside any selected corridor desert or deficit district, clicking a candidate site pin reveals a comprehensive **Three-in-One Site Intelligence Drawer**:

```
┌──────────────────────────────────────────────────────────────────────────────┐
│                 UNIFIED SITE INTELLIGENCE DRAWER: SITE #NH48-RAJ-042         │
├──────────────────────────────────────────────────────────────────────────────┤
│ OVERALL SUITABILITY: 88.4 / 100 [PRIORITY RANK: #1 - IMMEDIATE INTERVENTION] │
├──────────────────────────────────────────────────────────────────────────────┤
│                                                                              │
│ [PART A: WHY THIS SITE? (RADICAL EXPLAINABILITY)]                            │
│ • Fleet Adoption Pressure (35% weight): Score 92/100                         │
│   ➜ Bridges an 82 km charging desert between Behror and Kotputli.            │
│   ➜ Surrounding RTO district suffers an acute EVCR deficit of 210:1.         │
│ • Road Hierarchy & Traffic (25% weight): Score 95/100                        │
│   ➜ Situated on 6-lane NH-48 with heavy intercity passenger and freight traffic.
│ • Power Grid Proximity (25% weight): Score 86/100                            │
│   ➜ Nearest 33/11 kV Substation is 1.1 km away (Kotputli DISCOM feeder).     │
│   ➜ Line voltage drop within CEA 6% limit; feeder distance 1.1 km is well within the 5 k
│ • Brownfield Advantage (15% weight): Score 90/100                            │
│   ➜ Co-located with an operational HPCL Petrol Pump & 24/7 Food Court.       │
│   ➜ 0 days land acquisition delay; existing dedicated parking bays and 24/7 lighting.
│                                                                              │
│ [PART B: TURNKEY EQUIPMENT & CAPACITY PLANNING]                              │
│ • Recommended Fast DC Guns:     4 × 60 kW CCS-2 DC Fast Charging Guns        │
│ • Recommended Slow AC Backup:   2 × 22 kW Type-2 AC Plugs                    │
│ • Total Installed Capacity:     284 kW                                       │
│ • Recommended Sanctioned Load:  315 kVA Dedicated Distribution Transformer (DTR)
│ • Projected Daily Energy:       1,440 kWh / day                              │
│ • Projected Daily Throughput:   48 – 64 electric vehicles / day              │
│ • Average Top-Up Session:       35 – 45 mins (20% to 80% SoC)                │
│                                                                              │
│ [PART C: STATUTORY IMPACT]                                                   │
│ ➜ Reduces maximum corridor inter-charger gap from 82 km down to 24 km        │
│   (Fully satisfies statutory Ministry of Power 25 km spacing mandate).       │
└──────────────────────────────────────────────────────────────────────────────┘
```

5. Action Bar: PM E-DRIVE 2030 Policy Simulator & Tender Studio

Embedded at the bottom of the Agency portal for instant policy execution:

1. **PM E-DRIVE 2030 Growth Forecaster:**
2. Interactive slider forecasting vehicle growth and required charger additions year-by-year (2026–2030) using state-specific CAGRs.
3. **1-Click RFP & Tender Allocation Sheet (CSV/Excel):**

4. Instant export of structured tender schedules containing: `Site ID`, `GPS Coordinates`, `Nearest Highway Milestone`, `Priority Rank`, `Recommended Gun Count`, `Power Rating (kW)`, `Substation Distance (km)`, and `Brownfield Partner Name`.
5. **Executive Detailed Project Report (DPR PDF):**
6. One-click generation of a formal, executive-ready PDF document containing regional maps, scorecards, engineering specifications, and economic justifications for state and national ministries.

---

6. Key Product USPs

Product Feature	Urja-X Solution	Competitor / Status Quo Limitation
Dual-Persona Utility	Seamlessly serves both everyday EV drivers (route safety) and government agencies (capital planning).	Competitor tools are either pure driver apps or purely theoretical academic dashboards.
Unified Site Intelligence	Clubs Siting, Capacity Sizing, and Explainability into one turnkey site specification.	Existing tools output isolated pins without equipment sizing or grid feasibility context.
Corridor Desert Radar	Discretizes linear highway vectors into 5 km windows to pinpoint real-world range anxiety.	Others rely on district centroid circles that fail on linear intercity corridors.
Grid-Feasible Siting	Pre-evaluates proximity to $11\text{kV}/33\text{kV}$ substations using empirical electrical grid data.	Blind placement on roads lacking high-voltage grid access, causing massive commissioning delays.
Policy-to-Tender Workflow	One-click export of PM E-DRIVE compliant tender spreadsheets and executive DPRs.	Manual, months-long consulting cycles to translate GIS maps into actionable procurement tenders.